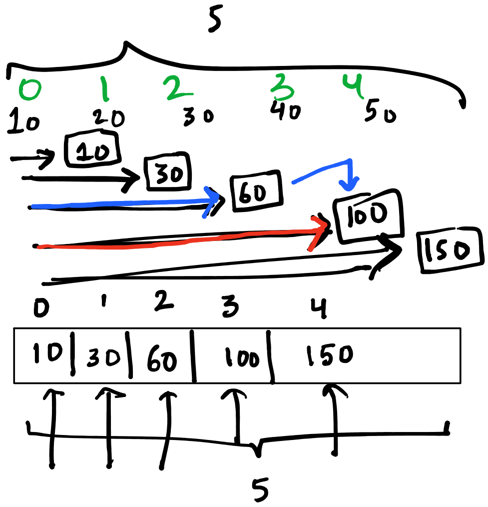
Prefix Sum Revision

Saturday, 14 June 2025 10:29 AM

$psum[3] = psum[2] + arr[3]$



⇒ psum[]

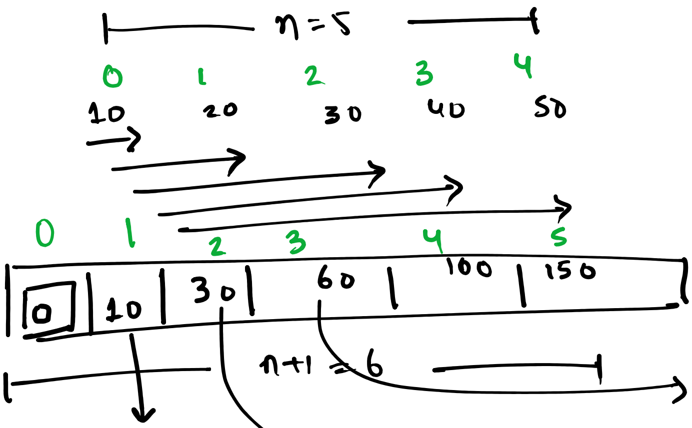
psum[i] stores the sum of prefix that ends at the i^{th} idx

$$\boxed{psum[i] = psum[i-1] + arr[i]}$$

$i \geq 1$

i=0

$$\boxed{psum_0 = arr_0}$$



psum'[]

sum of prefix that ends at idx 0

sum of prefix that ends at idx 1

sum of prefix that ends at idx 2

⇒ psum'[i] stores the sum of prefix that ends at the $(i-1)^{th}$ idx

$$\Rightarrow \boxed{psum[i] = psum[i-1] + arr[i-1]}$$

$$S_{ij} = \underbrace{\boxed{psum[j+1]}}_{\text{sum }(0 \text{ to } j)} - \underbrace{\boxed{psum[i]}}_{\text{sum }(0 \text{ to } i-1)}$$

sum(i to j)

$1e5 = 1 \cdot 10^5 = 10^5 \qquad n \le 1e5 \qquad q \le 1e5$

Difference Arrays

$Q_i: \; l_i, r_i, x_i \qquad [l_i, r_i]$

You are given an array of N integers, and Q queries each of the form { `li,ri,xi` } where `0<=li<=ri<=N-1`. Your task is to **apply** each of the Q queries to the given array & print the resulting array — applying a query to the array means adding `xi` to all the elements of the array with indexes from `li to ri` inclusive.

Example

N = 5, Q = 3

0	1	2	3	4
4	1	5	2	3

	l	r	x
Q_0	0	2	5
Q_1	2	3	2
Q_2	3	4	7

0 1 2 3 4

~~4~~ ~~1~~ ~~5~~ 2 3

9 6 ~~10~~ ~~2~~ 3

9 6 12 ~~4~~ ~~3~~

o/p: 9 6 12 11 10

$\boxed{l = 0 \qquad r = n-1 \qquad x}$

in the worst-case to apply a query we'll spend linear time assuming you iterate from l to r to apply a query

$\therefore$ total time : $O(qn)$

$10^5 \cdot 10^5 = 10^{10}$ ↓ TLE

$\le$ $\boxed{10^8 \sim 1s}$

n = 5

0	1	2	3	4
4	1	5	2	3

diff[l] += x;
diff[r+1] -= x; } const

q = 3

	l	r	x
Q_0	0	2	5
Q_1	2	3	2
Q_2	3	4	7

diff[]

0	1	2	3	4	5
~~0~~ 5	0	0	~~0~~ -5	0	0

n+1 = 6

extra slot to make impl. easier and avoid corner cases

diff[] 5 0 0 -5 0 0

linear

0	1	2	3	4	5
5	5	5	0	0	0

n = 5

0	1	2	3	4
~~4~~	~~1~~	~~5~~	~~2~~	~~3~~

9 6 ... 11 10

l r x

	l	r	x
Q_0	0	2	5

diff[l] += x
diff[r+1] -= x } const

	0	1	2	3	4
	~~4~~	~~1~~	~~5~~	~~2~~	~~3~~
	9	6	12	11	10

$diff[l] \mathrel{+}= x$
$diff[r+1] \mathrel{-}= x$ } const

$q = 3$

	l	r	x
Q_0	0	2	5
Q_1	2	3	2
Q_2	3	4	7

$n+1 = 6$

diff[]

0	1	2	3	4	5
~~0~~	0	~~0~~	~~0~~	~~0~~	~~0~~
5		2	~~-5~~ 2	-2	-7

$q \cdot const + n \cdot const + n \cdot const \therefore O(n+q)$

$diff[i] \mathrel{+}= diff[i-1]$

5 0 2 2 -2 -7

0	1	2	3	4	5
5	~~0~~	~~2~~	~~2~~	-2	-7
5	5	7	9	7	0

$\Rightarrow$ diff[]

0	1	2	3	4	5
5	5	7	9	7	0

Little Girl and Maximum Sum

The little girl loves the problems on array queries very much.

One day she came across a rather well-known problem: you've got an array of n elements (the elements of the array are indexed starting from 1); also, there are q queries, each one is defined by a pair of integers l_i, r_i $(1 \le l_i \le r_i \le n)$. You need to find for each query the sum of elements of the array with indexes from l_i to r_i, inclusive.

The little girl found the problem rather boring. She decided to reorder the array elements before replying to the queries in a way that makes the sum of query replies maximum possible. Your task is to find the value of this maximum sum.

Input

The first line contains two space-separated integers n $(1 \le n \le 2 \cdot 10^5)$ and q $(1 \le q \le 2 \cdot 10^5)$ — the number of elements in the array and the number of queries, correspondingly.

The next line contains n space-separated integers a_i $(1 \le a_i \le 2 \cdot 10^5)$ — the array elements.

Each of the following q lines contains two space-separated integers l_i and r_i $(1 \le l_i \le r_i \le n)$ — the i-th query.

Output

In a single line print, a single integer — the maximum sum of query replies after the array elements are reordered.

a[] [$\boxed{5}$ 3 2]

you want to maximize this

$2 \cdot 3 + 3 \cdot 5 + 2 \cdot 2$
$6 + 15 + 4$
[3, 5, 2] = 25
or
[2, 5, 3]

Examples

input
```
3 3
5 3 2
1 2
2 3
1 3
```
output
```
25
```

n = 3   q = 3
idx 1 2 3
a[] [5 3 2]
a_1 a_2 a_3

5+3 = 8
3+2 = 5
5+3+2 = 10
$\sum = \boxed{23}$

$a_1 + a_2$
$a_2 + a_3$
$a_1 + a_2 + a_3$
$\sum$

$2a_1 + 3a_2 + 2a_3$
2/3   5   2/3

input
```
5 3
5 2 4 1 3
1 5
2 3
2 3
```
output
```
33
```

a_1 a_2 a_3 a_4 a_5
n=5   q=3
[0 0 0 0 0]
1 2 3 4 5
a[] [$\boxed{5}$ 2 $\boxed{4}$ 1 3]

$a_1 + a_2 + a_3 + a_4 + a_5$
$a_2 + a_3$
$a_2 + a_3$

$1 \cdot a_1 + 3 \cdot a_2 + 3 \cdot a_3 + 1 \cdot a_4 + 1 \cdot a_5$ ← maximize this
[2 5 4 1 3]

diff [1 | 2 | 0 | -2 | 0 | -1]
1 2 3 4 5 6

$1 \cdot 2 + 3 \cdot 5 + 3 \cdot 4 + 1 \cdot 1 + 1 \cdot 3$
$= 2 + 15 + 12 + 1 + 3$
$= \boxed{33}$

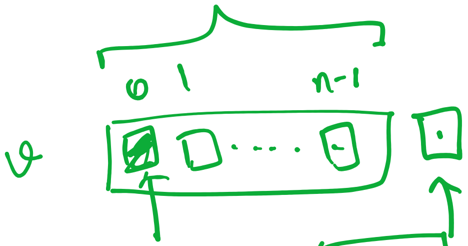
1 2 0 -2 0 -1
⇒ [1 3 3 1 1] [0]
a_1 a_2 a_3 a_4 a_5

[l, r] ⇒ diff[l] ++;
diff[r+1] --;

freq[ ] 1 3 3 1 1 $\Rightarrow$ 1 1 1 3 3

a[ ] 5 2 4 1 3 $\Rightarrow$ 1 2 3 4 5

$$\underbrace{1\cdot1 + 1\cdot2 + 1\cdot3 + 3\cdot4 + 3\cdot5}_{33}$$



sort ($\underbrace{\text{v.begin()}}_{[\text{first}}$, $\underbrace{\text{v.end()}}_{\text{last})}$)

[HW] Karen and Coffee

Karen and Coffee

Karen, a coffee aficionado, wants to know the optimal temperature for brewing the perfect cup of coffee. Indeed, she has spent some time reading several recipe books, including the universally acclaimed "The Art of the Covfefe".

She knows n coffee recipes. The i-th recipe suggests that coffee should be brewed between l_i and r_i degrees, inclusive, to achieve the optimal taste.

Karen thinks that a temperature is *admissible* if at least k recipes recommend it.

Karen has a rather fickle mind, and so she asks q questions. In each question, given that she only wants to prepare coffee with a temperature between a and b, inclusive, can you tell her how many admissible integer temperatures fall within the range?

Input

The first line of input contains three integers, n, k $(1 \le k \le n \le 200000)$, and q $(1 \le q \le 200000)$, the number of recipes, the minimum number of recipes a certain temperature must be recommended by to be admissible, and the number of questions Karen has, respectively.

The next n lines describe the recipes. Specifically, the i-th line among these contains two integers l_i and r_i $(1 \le l_i \le r_i \le 200000)$, describing that the i-th recipe suggests that the coffee be brewed between l_i and r_i degrees, inclusive.

The next q lines describe the questions. Each of these lines contains a and b, $(1 \le a \le b \le 200000)$, describing that she wants to know the number of admissible integer temperatures between a and b degrees, inclusive.

Output

For each question, output a single integer on a line by itself, the number of admissible integer temperatures between a and b degrees, inclusive.

Examples

input

```
3 2 4
91 94
92 97
97 99
92 94
93 97
95 96
90 100
```

output

```
3
3
0
4
```

input

```
2 1 1
1 1
200000 200000
90 100
```

output

```
0
```